

TASK 4

MongoDB Integration & SQL vs NoSQL Analysis

TechReads Data Engineering Pipeline

Comprehensive Technical Documentation

Learning Outcomes Covered:

- Implement NoSQL database integration
- Design and execute multi-database queries
- Create and optimize database indexes
- Compare SQL vs NoSQL performance and use cases
- Ensure idempotent data pipeline operations

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1. Introduction & Task Overview

Purpose: Task 4 evaluates your ability to extend the TechReads data pipeline from relational databases (MySQL) to NoSQL databases (MongoDB). This demonstrates understanding of modern polyglot persistence architectures where different database technologies are used for different workloads.

What is NoSQL?

NoSQL (Not Only SQL) refers to non-relational databases that store data differently than traditional SQL tables. Instead of rows and columns, NoSQL databases use flexible data structures like documents, key-value pairs, or graphs. MongoDB is a document-oriented NoSQL database where data is stored as JSON-like documents.

Why MongoDB for TechReads?

- 1. Flexible Schema:** Book data from different sources may have varying fields. MongoDB allows storing documents with different structures without requiring schema migrations.
- 2. Document Model:** A book with all its attributes (title, author, price, rating, URL, scrape timestamp) naturally maps to a MongoDB document.
- 3. Horizontal Scalability:** MongoDB can scale across multiple servers via sharding, ideal for large-scale data pipelines.
- 4. Idempotent Operations:** Upserts with unique keys ensure the pipeline is safe to re-run without duplicates.

Task 4 Components:

- 1. Data Loading:** Load CSV data into MongoDB with type transformations
- 2. Querying:** Implement 4 specific queries (Q1-Q4) matching brief requirements
- 3. Indexing:** Create indexes to optimize query performance
- 4. Performance Analysis:** Benchmark MongoDB vs MySQL with and without indexes
- 5. Comparative Study:** Evaluate SQL vs NoSQL strengths and trade-offs

2. MongoDB Setup & Connection

MongoDB Server Setup (macOS)

`brew services start mongodb-community`

This command starts MongoDB in the background on the default port 27017.

PyMongo Connection

PyMongo is the official Python driver for MongoDB. It allows us to connect to MongoDB and perform operations from Python code.

```
from pymongo import MongoClient, ASCENDING, DESCENDING
client = MongoClient('mongodb://localhost:27017/')
collection = client['techreads_db']['books']
```

Connection String Breakdown:

- **mongodb://** - Protocol identifier
- **localhost** - Host (local machine)
- **:27017** - Port (MongoDB default)
- **techreads_db** - Database name (created automatically if doesn't exist)
- **books** - Collection name (equivalent to table in SQL)

Connection Verification:

```
try:
    client = MongoClient('mongodb://localhost:27017/', serverSelectionTimeoutMS=5000)
    client.admin.command('ping')
    print('Connected to MongoDB')
except Exception as e:
    print(f'Connection failed: {e}')
```

3. Data Loading & Transformation

Overview

The CSV file created in Task 1 (techreads_books.csv) contains scraped book data. In Task 4, we load this data into MongoDB, transforming data types and ensuring deduplication through upsert operations.

Step 1: Load CSV File

```
import pandas as pd
df = pd.read_csv('data/techreads_books.csv')
print(f'Loaded {len(df)} books from CSV')
```

Step 2: Data Transformation

Before inserting into MongoDB, data types must be normalized:

CSV Field	Original Type	MongoDB Type	Transformation
year	Mixed string/int	Integer or Null	Parse as int; handle "N/A"
price	String with currency	Float	Remove symbols (£, \$, GBP)
star_rating	String/int	Integer 1-5	Convert to int
scraped_at	ISO string	String	Store as-is

Step 3: Upsert Documents

Upsert (update or insert) ensures that running the pipeline multiple times doesn't create duplicates. We use **book_url** as the unique identifier:

```
for _, row in df.iterrows():
    # Parse publication year
    year_raw = row.get('year', None)
    try:
        publication_year = int(year_raw) if year_raw and str(year_raw) != 'N/A' else None
    except (ValueError, TypeError):
        publication_year = None

    # Parse price (remove currency symbols)
    price_str = str(row['price']).replace('GBP', '').replace('£', '').replace('$', '')
    price = float(price_str.strip())

    # Create document
    doc = {
        'book_url': row['book_url'],
        'title': str(row['title']),
        'author': str(row['author']),
        'publication_year': publication_year,
        'price': price,
        'rating': int(row['star_rating']),
        'scraped_at': str(row['scraped_at'])
    }

    # Upsert: update if exists, insert if not
    collection.update_one(
        {'book_url': doc['book_url']}, # Filter: find by URL
```

```
{'\$set': doc}, # Update: set all fields  
upsert=True # Insert if not found  
)
```

Why Upsert is Important:

- **Idempotency:** If the pipeline runs twice, the second run updates existing documents rather than creating duplicates
- **Robustness:** Failed runs can be safely re-executed
- **Data Freshness:** New price or rating information is automatically updated

4. MongoDB Queries (Q1-Q4)

Four specific queries were required by the brief. These demonstrate different MongoDB query capabilities.

Query Setup

Define projection to select only needed fields:

```
PROJ = {'_id': 0, 'title': 1, 'author': 1, 'publication_year': 1, 'price': 1, 'rating': 1}
```

This returns 6 fields while excluding MongoDB's internal `_id` field.

Q1: Rating Filter (rating >= 4)

Purpose: Find highly-rated books

```
q1 = list(collection.find({'rating': {'$gte': 4}}, PROJ))
```

Explanation: The query operator `$gte` means 'greater than or equal'. This finds all documents where the rating field is `>= 4`.

Q2: Price Filter (price <= 20)

Purpose: Find affordable books under £20

```
q2 = list(collection.find({'price': {'$lte': 20.0}}, PROJ))
```

Explanation: The operator `$lte` means 'less than or equal'. This identifies budget-friendly books for cost-conscious readers.

Q3: Sorting (Top 10 Most Expensive Books)

Purpose: Identify premium-priced books for market analysis

```
q3 = list(collection.find({}, PROJ).sort('price', DESCENDING).limit(10))
```

Explanation:

- `find({})` - Match all documents (empty filter)
- `.sort('price', DESCENDING)` - Order by price from highest to lowest
- `.limit(10)` - Return only first 10 results

Q4: Year Filter (publication_year > 2018) - BRIEF REQUIREMENT

Purpose: Focus on recent Data Engineering titles (required by assessment brief)

```
q4 = list(collection.find({'publication_year': {'$gt': 2018}}, PROJ))
```

Explanation: The operator `$gt` means 'greater than'. This filters to books published after 2018, ensuring the dataset focuses on recent publications in this fast-evolving field.

5. Indexing Strategy (Learning Outcome 4)

Indexes are data structures that dramatically improve query performance by enabling fast data lookup. B-tree indexes are the standard approach used by both SQL and NoSQL databases.

Why Indexes Matter at Scale

- **Small dataset (20 books):** Full table scan takes ~0.1ms anyway
- **Large dataset (1M books):** Full scan takes ~1-5 seconds; indexed query takes ~5-10ms
- **Difference:** 1000x performance improvement at scale

Creating Strategic Indexes

```
# Drop old indexes to compare before/after
collection.drop_indexes()

# Create three targeted indexes
collection.create_index([('rating', ASCENDING)], name='idx_rating')
collection.create_index([('price', DESCENDING)], name='idx_price')
collection.create_index([('publication_year', ASCENDING)], name='idx_year')
```

Index Purpose & Optimization

Index Name	Field	Direction	Optimizes Query
idx_rating	rating	ASCENDING	Q1: Find ratings >= 4
idx_price	price	DESCENDING	Q3: Sort by price DESC, Q2: Filter price <= 20
idx_year	publication_year	ASCENDING	Q4: Filter year > 2018

Verify Indexes Created

```
for idx in collection.list_indexes():
    print(idx['name'], idx['key'])
```

B-tree Index Mechanism

Indexes organize data in a balanced tree structure. For a sorted field:

- **Without index:** Database must scan all ~1M documents ($O(n)$ complexity)
- **With index:** Database performs binary search in tree: 20 comparisons max ($O(\log n)$ complexity)
- **Trade-off:** Indexes require extra disk space and slow down writes slightly (maintain index on INSERT/UPDATE)

6. Performance Analysis: MongoDB vs MySQL

To validate our indexing strategy, we benchmarked the same four queries on both MongoDB and MySQL, measuring execution time before and after creating indexes.

Benchmarking Methodology

```
# Execute query 10 times and measure
times = []
for _ in range(10):
    t0 = time.perf_counter()
    results = list(collection.find(filter))
    times.append((time.perf_counter() - t0) * 1000) # Convert to ms

avg_time = sum(times) / len(times)
print(f'Average: {avg_time:.3f}ms')
```

Expected Results

WITHOUT Indexes (Full Table Scan):

- All queries take similar time (~1-5ms per query on small dataset)
- Database must examine every document

WITH Indexes:

- Filtered queries (Q1, Q2, Q4) become faster (~0.5-2ms)
- Sorted query (Q3) benefits from index on sort field
- At large scale, improvement is dramatic (100x-1000x)

MongoDB vs MySQL Comparison

Aspect	MongoDB	MySQL	Note
Connection	Native Python driver	mysql-connector-python	Both efficient on local machine
Query Speed	Similar to MySQL at scale	Similar to MongoDB at scale	Indexes determine performance
Complex Queries	Simpler (document-focused)	Better (JOIN support)	MongoDB uses aggregation pipeline
Schema Changes	Zero overhead	Requires ALTER TABLE	MongoDB advantage for evolution
ACID Support	Multi-document transactions	Full ACID	MySQL stronger for consistency

7. SQL vs NoSQL: Comprehensive Comparison

Data Model

SQL (MySQL): Structured as tables with fixed schema. Each row has identical columns. Relationships between tables via foreign keys.

NoSQL (MongoDB): Documents with flexible schema. Each document is self-contained. No rigid structure required.

Example: Storing a Book in Each Database

MySQL:
INSERT INTO books (title, author, price, rating)
VALUES ('Data Engineering', 'Jane Doe', 29.99, 5);

MongoDB:
{
 'title': 'Data Engineering',
 'author': 'Jane Doe',
 'price': 29.99,
 'rating': 5
}

Characteristic	SQL (MySQL)	NoSQL (MongoDB)	Winner for TechReads
Schema Flexibility	Fixed; ALTER TABLE needed	Dynamic; add fields anytime	MongoDB (evolving data)
Data Consistency	ACID (guaranteed)	Eventual consistency	MySQL (critical for sales)
Horizontal Scaling	Difficult (vertical only)	Built-in sharding	MongoDB (future growth)
Query Complexity	Excels with JOINS	Single documents best	MySQL (complex reports)
Write Speed	Slower (ACID overhead)	Faster (eventual consistency)	MongoDB (high volume)
Learning Curve	SQL standardized	Multiple query languages	SQL (mature standard)

Polyglot Persistence Recommendation

Modern data pipelines often use **both** SQL and NoSQL:

- **MySQL:** Authoritative source for financial data, reporting, complex analytics
- **MongoDB:** Cache layer for frequently accessed data, document storage, rapid ingestion
- **Result:** Best of both worlds - consistency + scalability

8. Real-World Application & Industry Context

When Companies Use MongoDB

1. **Content Platforms (Twitter, MongoDB):** Users can have different fields; schema must be flexible.
2. **E-commerce (Amazon, eBay):** Products have varying attributes; catalogs constantly evolve.
3. **Real-time Analytics (Uber, Airbnb):** Need fast writes and horizontal scaling across servers.
4. **IoT Systems:** Different sensors produce different data formats; centralize into documents.

When Companies Use MySQL/SQL

1. **Banking (transactions must never be lost):** ACID compliance is non-negotiable.
2. **ERP Systems:** Complex multi-table queries, strict referential integrity required.
3. **Business Intelligence:** Complex JOINS and aggregations across normalized tables.
4. **Financial Reporting:** Data consistency and audit trails are critical.

TechReads Use Case Analysis

TechReads benefits from BOTH databases because:

MySQL Stores: Canonical book records, pricing, inventory (consistency critical)

MongoDB Stores: Scraped metadata, reviews, tags (schema varies by source)

This architecture demonstrates enterprise data engineering practices where technology choice depends on workload.

9. Troubleshooting & Verification

Common Issues & Solutions

Issue 1: 'Connection refused' error

- Check MongoDB is running: `brew services list | grep mongod`
- Start if needed: `brew services start mongodb-community`

Issue 2: 'No module named pymongo'

- Install package: `pip install pymongo`
- Within venv: `source .venv/bin/activate` first

Issue 3: Type conversion errors on price/year

- Add try/except blocks as shown in the data loading section
- Set to None if parsing fails

Issue 4: Duplicates after re-running pipeline

- Ensure `upsert=True` is used in `update_one()`
- Verify `book_url` is unique and consistent

Verification Commands

```
# Check document count
collection.count_documents({})

# View sample document
collection.find_one()

# Count documents matching Q1 filter
collection.count_documents({'rating': {'$gte': 4}})

# List all indexes
list(collection.list_indexes())

# Check index statistics
collection.index_information()
```

10. Learning Outcomes & Summary

Learning Outcomes Achieved (LO4):

- ✓ **NoSQL Database Integration:** Successfully implemented MongoDB in the data pipeline alongside MySQL
- ✓ **Document-Oriented Design:** Modeled book records as MongoDB documents with appropriate transformations
- ✓ **Idempotent Operations:** Used upserts with unique keys to ensure pipeline can be safely re-run
- ✓ **Query Implementation:** Implemented all 4 required queries (Q1-Q4) with proper filtering and sorting
- ✓ **Indexing Strategy:** Created strategic B-tree indexes and understood performance implications
- ✓ **Performance Analysis:** Benchmarked before/after indexing on both SQL and NoSQL databases
- ✓ **Comparative Analysis:** Evaluated SQL vs NoSQL trade-offs with informed recommendations

Key Takeaways

1. **Polyglot Persistence:** Modern data engineering uses multiple database technologies for different purposes.
2. **Schema Flexibility:** NoSQL excels when data structures evolve; SQL excels when consistency is paramount.
3. **Performance Optimization:** Indexes are critical at scale; B-tree structure reduces $O(n)$ to $O(\log n)$ queries.
4. **Idempotency:** Production pipelines must be re-runnable; upserts with unique keys ensure this.
5. **Benchmarking:** Performance improvements are dataset-size dependent; test assumptions with real data.

Task 4 Completion Checklist

Requirement	Completed	Evidence
MongoDB integration setup	✓	Connection to localhost:27017
CSV data loaded with transformations	✓	Type conversions applied (int, float)
Upsert operations implemented	✓	book_url as unique key prevents duplicates
Q1: Rating ≥ 4 query	✓	Filter with \$gte operator
Q2: Price ≤ 20 query	✓	Filter with \$lte operator
Q3: Top 10 by price sort	✓	Sort descending, limit 10
Q4: Year > 2018 (brief requirement)	✓	Temporal filter on publication_year
Three indexes created	✓	idx_rating, idx_price, idx_year
Performance benchmarking	✓	Before/after timing measurements
SQL vs NoSQL analysis	✓	Comparison table with recommendations

Conclusion

Task 4 demonstrates a production-grade understanding of database technology evaluation and integration. By implementing both SQL and NoSQL solutions, benchmarking performance, and analyzing trade-offs, you have shown the decision-making skills required in real-world data engineering roles where technology selection depends on specific workload characteristics. The TechReads pipeline now spans the complete data engineering lifecycle: collection → transformation → relational storage → document storage → optimization.